EXPERIMENT NO. 11

Implement Interfaces and Multiple Inheritance through Interfaces

PART B – TO BE COMPLETED BY STUDENTS

Roll No. C-60

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Class / Division: SE-C

Batch: C2

Date of Experiment _____

Date of Submission _____

Grade _____

B.1 Document Created by the Student

Add a third interface and implement it in the same class.

A third interface named Recordable is added to represent the recording of patient monitoring data. The PatientMonitor class implements Monitorable, Alertable, and Recordable. This demonstrates multiple inheritance through interfaces in Java.

B.2 Student Program / Output

Paste/write the final program and attach or reproduce the observed output below.

```
interface Monitorable {
   void monitor();
}
interface Alertable {
   void sendAlert();
   default void systemMessage() {
      System.out.println("Alert system ready.");
   }
}
```

```
interface Recordable {
    void recordData();
}
class PatientMonitor
        implements Monitorable, Alertable, Recordable {
    public void monitor() {
        System.out.println("Patient monitoring started.");
    }
    public void sendAlert() {
        System.out.println("Alert sent to authorized staff.");
    }
    public void recordData() {
        System.out.println("Patient data recorded successfully.");
    }
}
public class InterfaceDemo {
    public static void main(String[] args) {
        PatientMonitor monitor = new PatientMonitor();
        monitor.systemMessage();
        monitor.monitor();
        monitor.sendAlert();
        monitor.recordData();
    }
}
```

```
Alert system ready.
Patient monitoring started.
Alert sent to authorized staff.
Patient data recorded successfully.
```

B.3 Observations and Learning

● Why are interfaces called contracts?

Interfaces are called contracts because they specify methods that implementing classes must provide, subject to the language's interface rules.

● How can Java achieve multiple inheritance using interfaces?

Java allows a class to implement multiple interfaces using the implements keyword. The class must implement the required abstract methods from those interfaces.

● What is a default method in an interface?

A default method is a method declared with the default keyword that includes an implementation inside an interface. Implementing classes can use it directly or override it.

Student observation / learning:

I learned how interfaces define contracts and how one class can implement multiple interfaces. The PatientMonitor class performs monitoring, sends alerts, and records patient data.

B.4 Conclusion

Smart wearable device interfaces are included as a syllabus self-learning context.

Student conclusion:

The experiment successfully demonstrates interfaces and multiple inheritance through interfaces in Java. The PatientMonitor class implements three interfaces and provides the required functionality, including the use of a default method.

B.5 Questions of Curiosity

6. Why are interfaces called contracts?

They define the methods that an implementing class is expected to provide.

7. How can Java support multiple inheritance through interfaces?

A Java class can implement two or more interfaces, combining their contracts without inheriting from multiple classes.

8. What is a default interface method?

A default interface method is a method with a body declared using the default keyword, allowing implementing classes to inherit its implementation.